Tiled Benchmark Analysis

- Testing with tile sizes of 8×8, 16×16, 32×32
  - Reasoning
    - 8×8 → 64 threads → 2 warps full
    - Above 32×32
      - 64×64 is 4096 threads/block
        - max is 1024 can't do
- Tested across matrix sizes 2048 & 4096
  - Established in naive kernel that workload was too small
  - Tiled kernel should run faster than naive so the noise from ramp-up and tail effects would be even worse

- 2048 × 2048
  - Best performing tile size: 32
    - cuBLAS is 33% faster
    - Most reuse compared to other tile sizes
      - 2× of 16 and 4× of 8
      - ½ global mem pulls of 16, ¼ of 32
    - Downside of bigger tile size
      - ↳ Block has more warps
        - ↳ Fewer blocks fit in 1 SM
          - ↳ SM has less blocks to switch to in order to latency hide
            - ↳ SM more idle
      - SM has 48 warp limit on 5080
        - 32 tile size → 32 warps/block → 1 block/sm
  - 2048 × 2048
    - Small enough that fits in L2
      - 64 mb L2 size
      - $2048^2 \times 4B \times 3$ matrices = 50.3mb
    - Memory misses don't cost that much so even though SM has fewer blocks it doesn't have to switch as much since there's less latency to hide

- 4096 x 4096
  - Best Performing Tile Size: 16
    - Balances amount of reuse with # of blocks
      - Crucially here all the data cannot fit in L2
        - $4096^2 \times 4B \times 3$ matrices = 201 MB > 64 MB
      - ↳ this means have blocks to switch to on the SM for latency hiding matters more
        - since more fetches from slower DRAM
    - cuBLAS is 98% faster

- Tile Size 8 was worst for all cases